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My future goal is to build robots that are ideally suited to the unique needs of my community, and that starts with understanding the very building blocks of those machines (semiconductors). To understand this goal, you need to know more about me. I'm from a small village in The Gambia called Tambasansang. I'm the seventh of ten kids, and I'm proud to say I was the first in my family to finish high school and go to college. My love for how machines work started when I was very young. I became a car mechanic's apprentice, and being around engines and electronics just captivated me. I truly fell in love with the world of machines. When I got to high school and learned about physics, math, and engineering, it felt like I'd found my calling. Then I wanted to be a mechanical engineer because I wanted to understand everything deeply about machines.

But life, as it often does, is unpredictable. In 2016, Libya was still in chaos, and a lot of people were taking advantage of that to cross the Mediterranean to Europe, especially Italy, looking for a way out. My family really wanted me to go too. At first, I said no. My eyes were fixed on finishing high school and getting into university to become a mechanical engineer. Then, one day, my sister said something that really hit home: if you genuinely wanted a higher education, you needed to leave the country for a place with a better system. And, honestly, we just couldn't afford it here. After thinking about her words for a couple of weeks, I decided to leave. I packed my bags, and off I went.

That journey changed me in remarkable ways. It was the most dangerous, dehumanizing, and heartbreaking thing I've ever experienced. Yet, strangely, it was also the most thrilling and adventurous. I nearly died from hunger and the brutal desert weather. I saw people do desperate things, including selling themselves just to survive another day. I witnessed shootings that crippled people, we had to bury a friend along the way, and I literally had a 9mm pistol pressed to my head, almost losing my life again, just because someone could take it if they wished it so.

When I finally crossed the Mediterranean and landed on Italian shores, my resolve to get a better education was stronger than ever. I threw myself into learning the language and enrolled in a technical high school. Mechanical engineering was still my dream, but then, almost by chance, I kept hearing about a place called United World College (UWC). I applied to the one in Italy and, to my amazement, got in. That school was a game-changer. It was there that I truly saw physics as the fundamental bedrock of all the engineering fields I was so passionate about. It completely shifted my perspective, and I decided to major in physics.

Fast forward to the summer of 2023, my freshman year at The College of Idaho. I was fortunate to work as an astronomy research assistant with Dr. Katie Devine on the “The Milky Way Bulge Extra-Tidal Star Survey (MWBest)” project, a role I maintained throughout my senior year. My initial work in this research in longslit spectroscopy made me an author in the “Automated Spectroscopic Wavelength Calibration using Dynamic Time Warping” paper. But in my opinion, my most significant achievement is developing a data reduction pipeline for the multi-slit spectroscopy method. While I absolutely love physics, this research, along with some of my lab work in classes, made me realize something important: I really love the hands-on, experimental side of it. That's when I knew I wanted to become an experimental physicist.

I remember one particular moment vividly during that summer research. Dr. Jim Dull, another one of my physics professors, had just returned from Italy and found my teammate and me working. He just popped over to say hi, and we ended up chatting for about half an hour. The only thing he talked about was units of measurement: centimeters, meters, miles, kilometers, inches, yards, and foot-longs. And it was honestly one of the best conversations of my life. After he left, my teammate and I just looked at each other, and all our motivation to work vanished; all we wanted was to keep talking to him. That evening, I asked myself: Is there any topic in the world I could talk about with that level of deep understanding? The answer was a clear "No." That's when it hit me: an undergraduate degree just isn't enough. I needed to attend graduate school to achieve that level of knowledge.

But getting to graduate school means being smart about my choices. While pursuing higher education is paramount, I also want to focus on something that lets me make a real difference in my community back home. I grew up in a farming village. My mom and dad couldn't read or write, and our entire livelihood depended on farming. It's incredibly physically demanding work, so much so that most young kids don't have time for school, let alone for dreaming and exploring other things that could help our community grow. For instance, my brother is an incredible soccer player who is known throughout our area, even in the cities. However, he couldn't focus because the family needed him; thus, he had to sacrifice his dream of going professional.

Imagine if my community had a farming robot that could handle most of the heavy labor. That would create ample space for kids to focus on their education and engage in creative activities that could genuinely bring about progress. While advanced farming equipment exists, the price tags are often more than my dad has ever owned in his entire life. Although my college doesn't have a robotics department, it gave me the freedom to explore my interests. I taught myself to use Fusion 360 (CAD software) (before college), as well as how to utilize microcontrollers like Arduino and Raspberry Pi, and ROS2 (a robotics operating system). These skills are letting me experiment with things that could truly benefit my community. Currently, I'm working on designing an affordable farming robot specifically for communities like mine.

While my primary goal is to build an institution that tackles the technological hardships communities like mine face, I've come to a critical realization. I can teach myself to assemble and program robots, but those machines rely on controllers, which are fundamentally built on semiconductor chips. My current understanding of the intricate physics and computational design behind these semiconductors is limited. To truly innovate rather than just piecing together existing, inadequate parts, I need to understand and optimize these fundamental building blocks. This requires a formal, in-depth education in computational modeling and simulation to bridge the gap between system-level robotics and deep semiconductor design.

Where I come from, we are often labeled a "third-world country," and the default approach is to simply copy what the West is doing. However, the technologies designed for the United States or Europe are rarely suited for the harsh, unique realities of our communities. To build agricultural robots ideally suited to Tambasansang, I must be able to design and optimize them from the atomic level up. Pursuing a Ph.D. in Computing with an emphasis in Computational Math Science and Engineering at Boise State University will provide the exact computational toolkit I need to achieve this.